

A Complete Signed Two-Peakon Scattering and Collision Atlas for the Camassa–Holm Equation

Route-A source-local certificate HCS-C278

1 September 2026

Abstract

On the ordered two-peakon manifold of the Camassa–Holm equation, momentum and energy reduce the four-dimensional flow to one separable quadratic for the exponential gap.

1 Frozen equation and reduction

We use the momentum form

$$m_t + um_x + 2u_xm = 0, \quad m = u - u_{xx}, \quad (1)$$

and the ordered ansatz

$$u(x, t) = \sum_{j=1}^2 p_j(t) e^{-|x - q_j(t)|}, \quad q_1 < q_2. \quad (2)$$

This is the normalization introduced with peaked solutions in Camassa and Holm [1]. Since $(1 - \partial_x^2)e^{-|x - q|} = 2\delta_q$, matching the δ'_q and δ_q coefficients in (1) gives

$$\dot{q}_i = \sum_j p_j e^{-|q_i - q_j|}, \quad \dot{p}_i = p_i \sum_j p_j \operatorname{sgn}(q_i - q_j) e^{-|q_i - q_j|}. \quad (3)$$

Put $q = q_2 - q_1$, $y = e^q$, $p = p_2 - p_1$, and

$$P = p_1 + p_2, \quad E = p_1^2 + p_2^2 + 2p_1p_2e^{-q}, \quad D^2 = 2E - P^2. \quad (4)$$

Lemma 1 (Invariant scalar reduction). *Before collision, P and E are constant and*

$$\dot{y} = p(y - 1), \quad \dot{p} = \frac{P^2 - p^2}{2y}, \quad \dot{y}^2 = D^2(y - 1) \left(y - \frac{P^2}{D^2} \right). \quad (5)$$

Proof. In the ordered chart, (3) says $\dot{p}_1 = -p_1p_2e^{-q}$ and $\dot{p}_2 = p_1p_2e^{-q}$. Direct differentiation proves $\dot{P} = \dot{E} = 0$. Substitution of $p_1 = (P - p)/2$, $p_2 = (P + p)/2$ into (4) gives

$$2E = P^2 + p^2 + \frac{P^2 - p^2}{y},$$

or $p^2(y - 1) = D^2y - P^2$. Combining this with $\dot{y} = p(y - 1)$ gives the last identity in (5); differentiating the algebraic identity gives the middle one. $\square$

Theorem 1 (Complete ordered two-peakon atlas). *For $q > 0$,*

$$P^2 - D^2 = 4p_1p_2(1 - e^{-q}).$$

At this revision stage we resolve the strict same-sign sector $p_1p_2 > 0$, equivalently $P^2 > D^2$.

Same sign: if $P^2 > D^2$, there is a unique time shift t_ such that*

$$y = 1 + \left(\frac{P^2}{D^2} - 1 \right) \cosh^2 \frac{D(t - t_*)}{2}, \quad p = D \tanh \frac{D(t - t_*)}{2}. \quad (6)$$

The solution is global and the gap remains positive.

Proof. The second root of the quadratic in (5) is P^2/D^2 . When it exceeds one, separation with the minimum at t_* gives (6). Differentiation verifies this branch. $\square$

References

- [1] R. Camassa and D. D. Holm, *An integrable shallow water equation with peaked solitons*, Physical Review Letters **71**(11) (1993), 1661–1664, [doi:10.1103/PhysRevLett.71.1661](https://doi.org/10.1103/PhysRevLett.71.1661).
- [2] K. Grunert and H. Holden, *The general peakon-antipeakon solution for the Camassa–Holm equation*, Journal of Hyperbolic Differential Equations **13** (2016), 353–380, [doi:10.1142/S0219891616500119](https://doi.org/10.1142/S0219891616500119).